

Mohamed Mahmoud Samy

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Portfolio & Projects: mohamedmsamy.github.io

SUMMARY

Third-year Mechatronics student at Ain Shams University with experience across embedded systems, PCB design, industrial automation, and autonomous robotics. I've interned at an aerospace agency and an automotive group, and I currently build autonomous race cars with F1TENTH. I like working on things that have to work in the real world.

EDUCATION

B.Sc. Mechatronics Engineering

Ain Shams University · Expected 2027 · GPA: 3.22 / 4.0

EXPERIENCE

Autonomous Systems Engineer | *F1TENTH Racing Team*

September 2025 – Present · Ain Shams University

- Developed and tuned autonomous navigation and control algorithms for a race vehicle using LiDAR-based perception data.
- Implemented Follow-The-Gap (FTG), Pure Pursuit, and Model Predictive Control (MPC) algorithms for obstacle avoidance and trajectory tracking.
- Collaborated with localization and perception sub-teams to integrate a real-time autonomous driving architecture in ROS2.

FPGA Design Intern | *EgSA – Egyptian Space Agency*

January 2025 – February 2025 · New Administrative Capital, Egypt

- Designed and implemented FPGA modules for embedded aerospace systems including finite state machines, hierarchical components, and arithmetic logic units in VHDL
- Performed RTL simulation, functional verification, and timing analysis to validate designs prior to hardware deployment.

Engineering Intern | *Ghabbour Auto*

July 2024 – August 2024 · Egypt

- Disassembled and inspected engine, manual transmission, and brake systems at the service facility.
- Performed precision measurements using dial gauges, cylinder bore gauges, straight edges, and Vernier calipers including crankshaft runout, engine block flatness, cylinder bore diameter, and brake drum diameter.

CERTIFICATIONS

Python for Everybody — University of Michigan (2026)

FPGA & VHDL Workshop — EgSA – Egyptian Space Agency (2025)

SOLIDWORKS 3D CAD for Education — Dassault Systèmes (2025)

MATLAB & Simulink: Onramp + Modeling & Simulation — MathWorks (2025)

OSHA Construction & Industry Course — Orascom Construction PLC (2025)

SKILLS

Embedded & Control: Embedded Systems, STM32, PLC, Control Systems, Electro-Pneumatics, ROS2

Electronics & Hardware: PCB Design, Circuit Simulation, Sensors & Actuators, FPGA / VHDL, Industrial Control Components

Mechanical Design: SolidWorks, CAD, Mechanical Drawings, Power Transmission, Stress Analysis

Programming & HDL: C / C++, Python, VHDL, MATLAB / Simulink